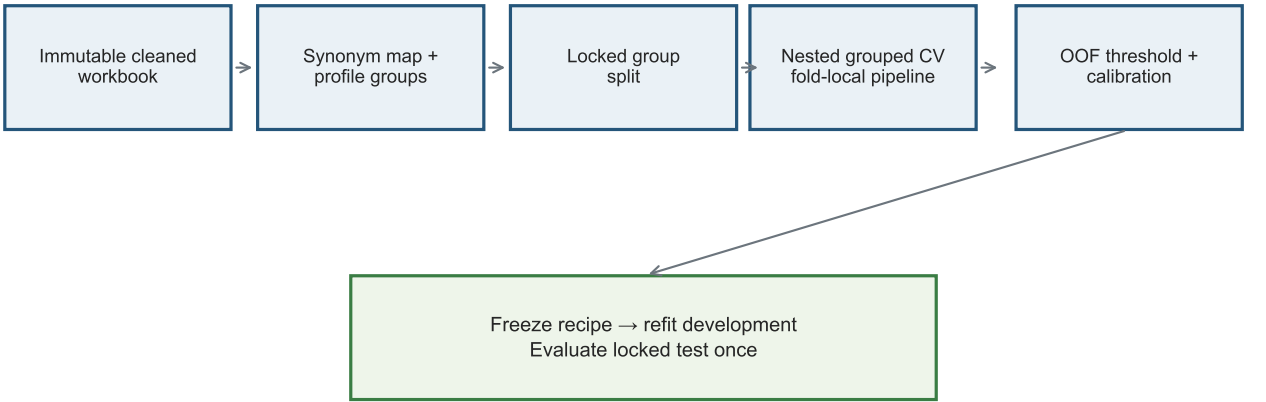
Immutable cleaned workbook
Synonym map + profile groups
Locked group split
Nested grouped CV fold-local pipeline
OOF threshold + calibration
Freeze recipe → refit development
Evaluate locked test once